

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) • Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) • Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions • Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians)	4
01.2	• Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) • Islands provide roosting / nesting sites safe from terrestrial predators (foxes) • Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks	3
01.3	• Mobile species (birds, flying insects) colonise easily • Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially	2
01.4	• Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	• Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere • Allowing more plant growth, further stabilising climate and soils	3
02.2	<i>Any 3 from:</i> • Lack of ancient historical / instrumental data • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions • Lack of sophisticated equipment / inability to measure many factors • Limited coordination between researchers; lack of data collection in many areas (oceans, poles)	3
02.3	• The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	
03.1	• Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced • Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares	2
03.2	• A hectare of land / sea with the world-average biological productivity • Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared	2
03.3	• $8.1 / 1.7 = 4.8$ (accept 4.7-4.8)	1
03.4	• Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint • Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) • Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016)	3
03.5	<i>Any 2 from:</i> • Renewable energy and energy conservation (carbon footprint is the largest component) • Reduced meat consumption / food waste; local seasonal food • Circular economy: repair, re-use, recycling; longer product lives • Public transport, efficient buildings	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	

Question	Marking guidance	Marks
04.1	• Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) • Fish are ectotherms so no energy is used for thermoregulation and water supports their mass; poultry grow fast and are kept warm indoors with movement restricted • Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion	3
04.2	• Control of movement (housing / restricted grazing) so less energy is used in muscle activity • Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature • Species selection: choose species with low FCRs (poultry, fish); high-energy easily digested diets; slaughter at the fastest growth stage	3
04.3	• Only about 10% of the energy at one trophic level is passed to the next; most is lost as heat in respiration / metabolism or in undigested material • Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) • Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot • Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	• Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned • Loss of canopy and litter exposes soil to rain splash and wind • Decomposing roots no longer bind soil so erosion, gully and landslides increase (especially on slopes) • Reduced organic matter and biota lower infiltration and water retention	4
05.2	• Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers • Increased albedo of cleared land (reflects more sunlight) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change	4
05.3	<i>Any 2 from:</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines • Loss of genetic resources / potential medicines before they are discovered	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
06.1	• Sigmoidal (S-shaped) growth curve • Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate • Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate • Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate	4
06.2	• Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation • Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter	2
06.3	• Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density • At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland)	2
06.4	• Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population • Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	
07.1	• Other MEDCs (USA, Canada, Australia) also had surpluses; Japan already had suppliers • Eastern Europe / USSR needed food but could not pay • Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production	3
07.2	• Quotas limiting production (eg milk) • Set-aside payments for taking land out of production • Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees) • Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming	4
07.3	<i>Any 3 from:</i> • Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape • Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat • Beetle banks of tussocky grass - natural pest predators • Skylark nesting plots; wild bird seed / nectar plants - winter food and pollinators • In-field tree protection; controlled stocking density to reduce compaction; ditch management; dry-stone walls; woodland management	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4.	

Question	Marking guidance	Marks
08.1	- The processes (eg regulation of atmospheric gases by photosynthesis and respiration) act in a dynamic equilibrium and cancel each other out - Because there is no overall change their actions go unnoticed / their importance is not appreciated until they are lost	2
08.2	<i>Any 3 from:</i> - Invertebrates, fungi and bacteria break down / decompose dead organic matter, releasing nutrients for plant growth - The organic matter and humus produced hold the soil together (reducing erosion) and regulate the water cycle, giving more even river flow and less flooding - Organic acids from decomposition aid the weathering of rock, producing more soil and releasing more nutrients - Microbes carry out the processes of the carbon, nitrogen and phosphorus cycles so wastes do not build up and nutrients are not depleted	3
08.3	- Trees reduce wind velocity so wind pollination would be ineffective / unreliable over long distances - Insects search over long distances for flowers so dispersed populations can be pollinated, and less pollen needs to be produced (energy saving); specialised flowers reduce pollen being wasted on other species	2
08.4	- Dispersal is not haphazard: animals live in the same habitat as the plant so are likely to carry seeds to locations where they can survive - Seeds can be carried longer distances than by wind and can be larger, containing more nutrients for the seedling - Seeds passing through the gut are deposited in faecal material which acts as a fertiliser, and passage through the intestine may stimulate germination	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.	
09.1	Waste products that are the raw materials for other processes	1
09.2	- Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) - The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2
09.3	- Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials - Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Protected areas: Korup NP Cameroon (1260 km², 600 tree species, endemic <i>Ancistrocladus korupensis</i> with anti-HIV / antimalarial alkaloids); Tumucumaque NP Brazil (39 000 km², adjoining French Guiana reserve); Alto Maues; Rio Bravo Belize funded by charities, companies and ecotourism • Debt for nature swaps; sustainable exploitation: selective logging of high-value timber (mahogany) funding conservation; ITTO aims (largely unmet); CITES listing of timber species (Honduras mahogany Appendix II) • Ecotourism income (but visitor damage in Costa Rica, Malaysia); community projects and employment; carbon sequestration payments • Effectiveness: protected areas work where enforced but face illegal logging, mining, poaching, roads, conflict; drivers (population growth, demand for beef, soya, palm oil, timber, minerals) originate outside the forest; climate change cannot be controlled locally • Judgement: combination of large protected areas, economic alternatives for local people and reduced international demand needed; overall loss continues	9
10.2	• Carbon sequestration reducing climate change; genetic resources / medicines; rainfall regulation; global biodiversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 56-57; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25